

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce

Department of Information Technology (B.Sc.I.T Semester III)

Python for Data Science

Practical – 1

Problem Statement

Roll No.:S056	Name:Shweta.Mohan.Singh
Class:SYIT	Batch:B1
Date of Assignment:15/7/2026	Date/Time of Submission: 15/7/2026

Q1A farmer wants a program to perform basic calculations related to farming.

The program should:

- Display the farmer's profile.
- Add the weights of two harvested crops.
- Find the square root of the total harvested area.
- Calculate the area of a triangular field.
- Swap the quantities of two fertilizer bags if entered in the wrong order.

```
import math

# 1. Display the farmer's profile
name = input("Enter farmer's name: ")
village = input("Enter village: ")
farm_size = float(input("Enter farm size (in acres): "))

print("\n--- Farmer's Profile ---")
print("Name:", name)
print("Village:", village)
print("Farm Size:", farm_size, "acres")

# 2. Add the weights of two harvested crops
crop1 = float(input("\nEnter weight of first harvested crop (kg): "))
crop2 = float(input("Enter weight of second harvested crop (kg): "))

total_weight = crop1 + crop2
print("Total Harvested Crop Weight:", total_weight, "kg")

# 3. Find the square root of the total harvested area
area = float(input("\nEnter total harvested area: "))
sqrt_area = math.sqrt(area)

print("Square Root of Harvested Area:", sqrt_area)

# 4. Calculate the area of a triangular field
base = float(input("\nEnter the base of the triangular field: "))
height = float(input("Enter the height of the triangular field: "))

triangle_area = 0.5 * base * height
print("Area of the Triangular Field:", triangle_area)

# 5. Swap the quantities of two fertilizer bags
bag1 = float(input("\nEnter quantity of Fertilizer Bag 1 (kg): "))
bag2 = float(input("Enter quantity of Fertilizer Bag 2 (kg): "))

print("Before Swapping:")
print("Bag 1:", bag1, "kg")
print("Bag 2:", bag2, "kg")

# Swapping
bag1, bag2 = bag2, bag1

print("After Swapping:")
print("Bag 1:", bag1, "kg")
print("Bag 2:", bag2, "kg")

print("Shweta S056")
```

Output:-

```
===== RESTART: C:/Users/itlab/OneDrive/D
Enter farmer's name: Raju
Enter village: Rampur
Enter farm size (in acres): 100

--- Farmer's Profile ---
Name: Raju
Village: Rampur
Farm Size: 100.0 acres

Enter weight of first harvested crop (kg): 120
Enter weight of second harvested crop (kg): 156
Total Harvested Crop Weight: 276.0 kg

Enter total harvested area: 300
Square Root of Harvested Area: 17.320508075688775

Enter the base of the triangular field: 240
Enter the height of the triangular field: 348
Area of the Triangular Field: 41760.0

Enter quantity of Fertilizer Bag 1 (kg): 120
Enter quantity of Fertilizer Bag 2 (kg): 150
Before Swapping:
Bag 1: 120.0 kg
Bag 2: 150.0 kg
After Swapping:
Bag 1: 150.0 kg
Bag 2: 120.0 kg
Shweta S056
```